

Compute

Hypervisors

Host Aggregates

Instances

Flavors

Images

Volume

Network

System

Reservation

Identity

Workflow

NFV

Project Name =

Filter

Delete Instances

Displaying 2 items

☐	Project	Host	Name	Image Name	IP Address	Flavor	Status	Task	Power State	Age	Actions	
☐	nfv	astha-VirtualBox	op-4fa4-8d22-1a5121bb3e14-RES PAWN-3-VDU1-57dwd52g44d5	Open WRT	10.0.0.173	openwrt_78127b9-69d4-4fa4-8d22-1a5121bb3e14-RESPAWN-3-VDU1_flavor-7qrcitlv3	Active		None	Running	0 minutes	Rescue Instance
☐	nfv	astha-VirtualBox	op-4427-b9fa-31ed6a0ded58-RES PAWN-1-VDU1-1eqc4lktpsmk	Open WRT	10.0.0.61, 10.0.0.100, 10.0.0.138	openwrt_fbfd3b42-1ff2-4427-b9fa-31ed6a0ded58-RESPAWN-1-VDU1_flavor-kap4ogodirect	Active		None	Running	14 minutes	Rescue Instance

Displaying 2 items

Figure 7: VNF instances

the project, the name of the VNF, the flavour used, and whether the instance is running or shut down.

A sample of the launched instance is shown in Figure 8.

Service function chaining (SFC): A feature of SDN

Traditionally, network configurations have hard-coded the routing of traffic through network services like load balancers and switches. This means that network managers must manually adjust the devices and interfaces to change the way traffic is routed through the network, making it difficult to adapt the network to changing requirements quickly. To address this issue, OpenStack has introduced service function chaining (SFC), which is essentially the software defined networking (SDN) version of policy-based routing (PBR).

SFC allows network managers to specify the order in which network functions, such as load balancers, firewalls, and VPNs, should be applied to incoming traffic. This approach is commonly used in NFV setups, where network operations are performed in software and run on standard hardware. With SFC, administrators can define a series of network operations that should

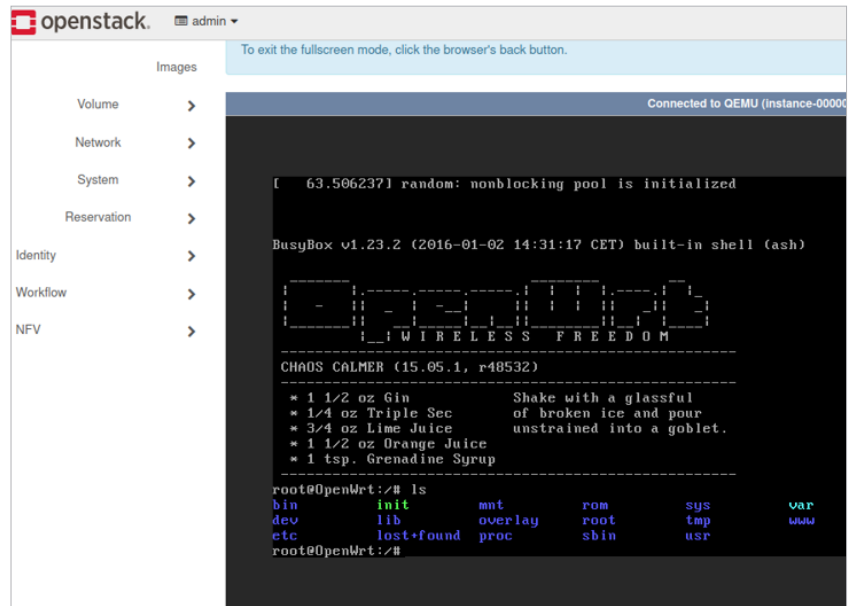


Figure 8: Launched VNF in OpenStack

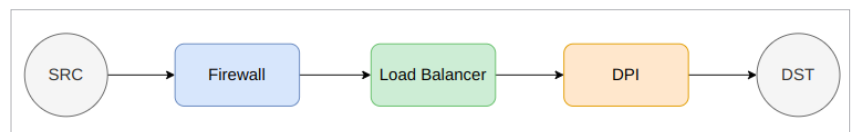


Figure 9: Chaining VNFs with different services

be applied to incoming traffic, ensuring that traffic is routed through its intended paths.

For example, in an SFC configuration, inbound traffic can be routed through a firewall for security screening before being

directed to a load balancer to be distributed among multiple servers. The traffic is then passed through these processes in the appropriate order, enhancing the network's flexibility and programmability. The sequence of VNFs connecting the source and destination can be seen in Figure 9.